

AI & MACHINE LEARNING

Complete Student Notes • Concepts • Algorithms • Examples • Exam Revision

Artificial Intelligence (AI)	A field of computer science concerned with building systems that perform tasks associated with human intelligence.
Machine Learning (ML)	A subset of AI in which models learn patterns from data for prediction or decision-making.
Deep Learning (DL)	A subset of ML based mainly on multi-layer neural networks, widely used for images, audio and language.

1. Artificial Intelligence - Introduction

Definition: AI develops computer systems capable of reasoning, learning, problem solving, perception, language understanding and decision making.

Goals of AI

- Build intelligent systems that solve real-world problems.
- Automate repetitive or complex tasks.
- Learn from experience and improve performance.
- Understand language, images and speech.

Applications

- Healthcare – diagnosis support and medical image analysis
- Agriculture – crop monitoring and disease detection
- Education – adaptive learning and AI tutors
- Banking – fraud detection
- Transportation – route planning and driver assistance
- E-commerce – recommendations
- Cybersecurity – anomaly detection

2. Machine Learning - Fundamentals

Machine Learning: Instead of programming every rule manually, we provide data and a learning method so a model can discover useful patterns.

Basic ML workflow

Problem Definition → Data Collection → Data Cleaning → Data Exploration → Feature Engineering → Train/Test Split → Model Training → Evaluation → Tuning → Deployment → Monitoring

Important terms

- Dataset: Collection of observations.
- Feature: Input variable.
- Label/Target: Desired output.
- Model: Learned representation of patterns.
- Training: Learning from training data.
- Inference: Prediction on new data.

3. Types of Machine Learning

Supervised Learning

Learns from labelled examples. Main tasks are classification (category prediction) and regression (numeric prediction).

Unsupervised Learning

Works with unlabelled data to discover structure, such as clusters or lower-dimensional representations.

Semi-Supervised Learning

Uses a small amount of labelled data together with a larger amount of unlabelled data.

Reinforcement Learning

An agent interacts with an environment and learns actions through rewards and penalties.

4. Data Preprocessing

Good preparation is essential because poor-quality input can produce unreliable models.

Common steps

1. Handle missing values.
2. Remove duplicates and obvious errors.
3. Encode categorical variables.
4. Scale numerical features when appropriate.
5. Inspect outliers.
6. Split data before fitting transformations to avoid data leakage.

Encoding: Label encoding maps categories to numbers; one-hot encoding creates separate binary columns.

Feature Scaling: Standardization commonly uses $z = (x - \mu) / \sigma$. Min-max normalization maps values to a selected range, often 0–1.

5. Train, Validation and Test Data

- Training set: Learns model parameters.
- Validation set: Helps tune and select models.
- Test set: Final evaluation on unseen data.

Data leakage: Information from outside the proper training process improperly influences the model, giving overly optimistic results.

6. Regression Algorithms

Linear Regression

Models a continuous target as a weighted combination of inputs. Simple form: $y = \beta_0 + \beta_1 x$.

Multiple Linear Regression

Uses several inputs: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$.

Decision Tree Regression

Splits the feature space into regions and predicts a value for each region.

Evaluation metrics

MAE, MSE, RMSE and R^2 .

7. Classification Algorithms

Logistic Regression

A classification method that commonly uses the sigmoid function to map a score to a value between 0 and 1.

Decision Tree

Uses a sequence of feature-based decisions. Easy to interpret, but can overfit.

Random Forest

An ensemble of randomized decision trees whose predictions are aggregated for better generalization.

K-Nearest Neighbors (KNN)

Predicts using nearby training examples. Feature scaling is often important.

Support Vector Machine (SVM)

Finds a decision boundary with a large margin; kernels can model nonlinear boundaries.

8. Clustering

K-Means Clustering

K-Means divides observations into K clusters by repeatedly assigning points to the nearest centroid and updating the centroids.

Steps: Choose K → initialize centroids → assign points → recompute centroids → repeat until stable.

Uses: customer segmentation, grouping documents, image compression and exploratory analysis.

Limitation: Choice of K matters and results can be affected by scaling and outliers.

9. Model Evaluation

Confusion Matrix

TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

F1-score = $2 \times (Precision \times Recall) / (Precision + Recall)$

For imbalanced classes: Accuracy alone may be misleading; precision, recall, F1-score and class-specific measures can be more informative.

10. Overfitting and Underfitting

Overfitting

The model learns training data too closely, including noise, and performs poorly on unseen data.

Underfitting

The model is too simple to capture important patterns and performs poorly even on training data.

Ways to reduce overfitting

Use more representative data, simplify the model, regularize, use cross-validation, tune hyperparameters, prune trees or use suitable early stopping.

11. Cross-Validation

K-Fold Cross-Validation: Divide training data into K folds. Train on K-1 folds and validate on the remaining fold; repeat so every fold is used for validation, then average the results.

Benefit: Helps estimate generalization performance and compare models more reliably when data is limited.

12. Hyperparameters

Hyperparameters are settings selected before or during training rather than learned directly as model parameters.

Examples: K in K-Means/KNN, tree depth, learning rate, number of trees, regularization strength, batch size and number of epochs.

13. Neural Networks

A neural network contains interconnected layers. A typical network has an input layer, one or more hidden layers and an output layer.

Neuron: $z = w_1x_1 + w_2x_2 + \dots + b$, followed by an activation function.

Common activation functions

- ReLU: $\max(0, x)$ – widely used in hidden layers.
- Sigmoid: output between 0 and 1 – often useful for binary probabilities.
- Softmax: converts scores into a probability distribution over multiple classes.

Training: Forward pass → calculate loss → backpropagation → update weights using an optimizer → repeat over batches/epochs.

Loss function: Measures the difference between predictions and targets. Examples include mean squared error and cross-entropy.

14. Deep Learning

Deep learning uses neural networks with multiple layers to learn increasingly useful representations from data.

CNN (Convolutional Neural Network)

Commonly used for images; convolutional layers learn local visual patterns such as edges and textures.

Sequence Models

Designed for sequential information such as text and time series. Modern systems also use Transformer architectures.

Transfer Learning

A pretrained model can be adapted to a related task, often reducing task-specific data and training requirements.

15. Natural Language Processing (NLP)

NLP enables computers to process and generate human language.

Applications: sentiment analysis, translation, text classification, information extraction, question answering, summarization and conversational systems.

Basic pipeline: Collect text → clean/normalize → tokenize → represent text → train/evaluate → deploy.

16. AI Ethics and Responsible AI

Important concerns include fairness, privacy, security, transparency, accountability, robustness and appropriate human oversight.

A good AI system should be evaluated not only for accuracy but also for how it behaves across relevant groups and real-world conditions.

17. AI vs ML vs Deep Learning

Aspect	AI	Machine Learning	Deep Learning
Scope	Broad field	Subset of AI	Subset of ML
Main idea	Intelligent behavior	Learn patterns from data	Learn representations with deep neural networks
Examples	Planning, reasoning, agents	Regression, classification, clustering	CNNs, deep sequence models, Transformers

18. Important Exam Questions

1. Define Artificial Intelligence and explain its applications.
2. Define Machine Learning and explain its types.
3. Differentiate classification and regression with examples.
4. Explain the ML workflow and data preprocessing.
5. Explain Linear Regression and Logistic Regression.
6. Explain Decision Tree and Random Forest.
7. Explain K-Means clustering and its steps.
8. Explain confusion matrix, Accuracy, Precision, Recall and F1-score.
9. Explain overfitting and underfitting.
10. Explain K-Fold Cross-Validation.
11. Explain neural networks, activation functions and backpropagation.
12. Explain Deep Learning, CNN and Transfer Learning.
13. Explain NLP and its applications.
14. Explain ethical issues in AI.

19. Quick Revision Sheet

AI → Intelligent systems
ML → Learning from data
Supervised → Labelled data
Unsupervised → Unlabelled data
Classification → Category prediction
Regression → Numeric prediction
Clustering → Group similar data
Overfitting → Good training performance, weak generalization
Underfitting → Model too simple
CNN → Common for image data
NLP → Human language processing
Epoch → One complete pass through training data
Feature → Input variable
Label → Target output

20. Short Viva Questions

- What is AI?
- What is ML?
- Difference between AI and ML?
- What is a feature?
- What is a label?
- What is training data?
- What is classification?
- What is regression?
- What is clustering?
- What is overfitting?
- What is a confusion matrix?
- What are precision and recall?
- What is an epoch?
- What is a neural network?
- What is deep learning?
- What is NLP?

Study Tip: For exams, write the definition first, then working/steps, followed by advantages, limitations and applications wherever relevant.